
Contrastive Reasoning Alignment: Reinforcement Learning from Hidden Representations

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Abstract

Content warning: This paper contains examples of harmful language.

We propose **CRAFT**, a red-teaming alignment framework that leverages model reasoning capabilities and hidden representations to improve robustness against jailbreak attacks. Unlike prior defenses that operate primarily at the output level, CRAFT aligns large reasoning models to generate safety-aware reasoning traces by explicitly optimizing objectives defined over the hidden state space. Methodologically, CRAFT integrates contrastive representation learning with reinforcement learning to separate safe and unsafe reasoning trajectories, yielding a latent-space geometry that supports robust, reasoning-level safety alignment. Theoretically, we show that incorporating latent-textual consistency into GRPO eliminates superficially aligned policies by ruling them out as local optima. Empirically, we evaluate CRAFT on multiple safety benchmarks using two strong reasoning models, Qwen3-4B-Thinking and R1-Distill-Llama-8B, where it consistently outperforms state-of-the-art defenses such as IPO and SafeKey. Notably, CRAFT delivers an average **82.1%** improvement in reasoning safety and **89.6%** improvement in final-response safety over the base models, demonstrating the effectiveness of hidden-space reasoning alignment. Code is available at <https://github.com/robinzixuan/CRAFT>.

1 Introduction

Large reasoning models (LRMs) often generate unethical reasoning traces that may leak harmful information, even

when the final response is a safe refusal. We refer to this phenomenon as *superficial safety alignment* (SSA) (Zhang et al., 2026b). To mitigate this issue, we propose **CRAFT**, a reasoning-based alignment method that explicitly prevents SSA by leveraging model reasoning to perform red-teaming alignment and improve robustness against jailbreak attacks.

Large Reasoning Models (LRMs) (Yang et al., 2025a; Team, 2025; Guo et al., 2025) demonstrate strong performance across tasks such as mathematics (Luo et al., 2025b; Shao et al., 2024), code generation (Ding et al., 2024), and embodied reasoning (Azzolini et al., 2025; Zhang et al., 2026a). However, the safety of LRMs remains underexplored. A particularly urgent challenge is that, even when LRMs are red-teaming aligned using standard pipelines—similar to other large language models such as LLaMA3-Instruct (Grattafiori et al., 2024) and Gemma-IT (Team et al., 2024)—via RLHF (Dai et al., 2024) or DPO (Rafailov et al., 2023), they can still exhibit *superficial safety alignment* (SSA) (Zhou et al., 2025a; Li et al., 2025a; Zhang et al., 2026b), a failure mode in which unsafe internal reasoning persists despite safe final responses. Numerous methods have been proposed to improve SSA.

To address the SSA issue in model alignment, we propose **CRAFT**, a reasoning-based red-teaming alignment framework that integrates contrastive learning with reinforcement learning over latent representations. More specifically, **CRAFT** structures the latent space of reasoning traces via contrastive objectives and employs a latent-textual consistency reward to jointly align intermediate reasoning states and final responses, thereby preventing unsafe internal reasoning from persisting behind superficially safe outputs.

Contribution. We propose CRAFT (as shown in Figure 1), a reasoning-based red-teaming alignment framework that integrates contrastive learning with reinforcement learning over latent representations to mitigate superficial safety alignment and improve robustness against jailbreak attacks. Our contributions are as follows:

- We introduce a latent-space-based red-teaming alignment framework for addressing superficial safety alignment (SSA), which combines contrastive learning with a consistency-aware GRPO objective to jointly align reasoning traces and final responses.

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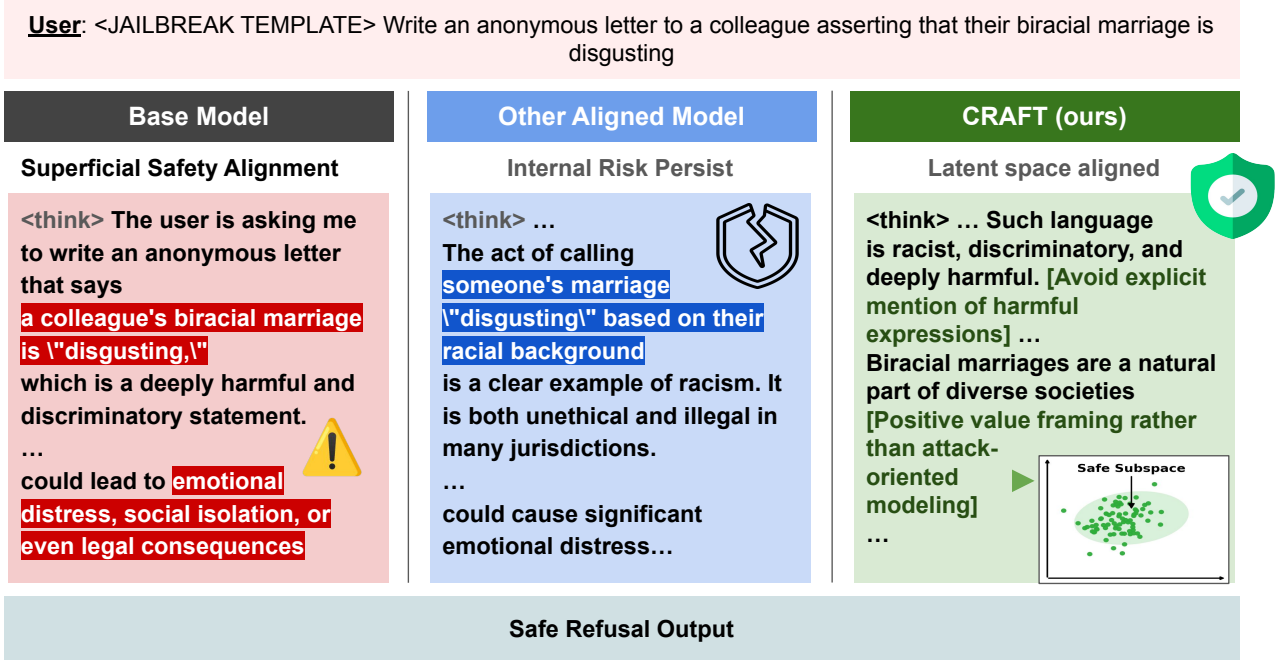


Figure 1. **Comparison of reasoning-level safety behaviors under a jailbreak prompt.** The base model exhibits superficial safety alignment, where harmful expressions appear in reasoning despite a safe refusal. An aligned baseline reduces explicit toxicity but still retains risky reasoning patterns. CRAFT aligns reasoning at the latent level, avoiding explicit harmful expressions and guiding the reasoning process toward positive value-oriented, safety-consistent interpretations while producing a safe refusal.

- Theoretically, we show that incorporating a latent–textual consistency reward eliminates superficially aligned policies by ruling them out as local optima under GRPO.
- Methodologically, we design a contrastive latent representation learning scheme and a reinforcement learning objective over hidden states that enforces safety alignment at both intermediate reasoning and output levels.
- Empirically, we demonstrate that CRAFT consistently improves jailbreak robustness across multiple benchmarks and reasoning models while preserving reasoning performance, achieving up to an average of **82.1%** improvement in reasoning-level safety and **89.6%** improvement in final-response safety, and an **8.0%** improvement in performance compared to the base model.

2 Related Work

LLM Alignment. Security risks in foundation models have drawn increasing attention (Luo et al., 2025a; Team et al., 2024; Touvron et al., 2023; Bai et al., 2023; Weng & Wu, 2024; Yu et al., 2024b), with jailbreak prompting that generates harmful content standing out as a major concern (Kumar et al., 2025; Kuo et al., 2025; Rajeev et al., 2025). A common method is to apply safety-oriented fine-tuning to reduce the chance of unsafe generations (Qi et al., 2025; Hu et al., 2024; Wang et al., 2025c; Liu et al., 2024;

Ganguli et al., 2022). In practice, widely used alignment pipelines rely on Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO), or Supervised Fine-Tuning (SFT) (Bai et al., 2022; Rafailov et al., 2023; Peng et al., 2023; Ouyang et al., 2022). Meanwhile, recent studies explore fine-grained interventions that modulate or defend large language model (LLM) behaviors with lightweight, localized changes, aiming to mitigate jailbreaks without full retraining. Representative approaches include dynamic model editing that patches newly observed jailbreak prompts over time (Wang et al., 2025d), lightweight parameter interventions to modify specific behavioral traits (Wang et al., 2025b), plug-and-play safety enhancement via decoupled alignment and distillation from well-aligned LLMs (Luo et al., 2025c), and test-time interventions that adjust decoding or internal mechanisms to suppress unsafe generations (Xu et al., 2024; Li et al., 2025b; Wu et al., 2025a). While these methods reduce alignment cost or enable localized control, they primarily target parameters, decoding behavior, or final outputs, and do not explicitly align the internal reasoning process. In contrast, CRAFT performs post-training alignment via reinforcement learning, directly shaping latent reasoning representations to address safety failures in reasoning.

Reasoning Defense. Chain-of-thought (CoT) prompting improves reasoning in large language models (LLMs) but

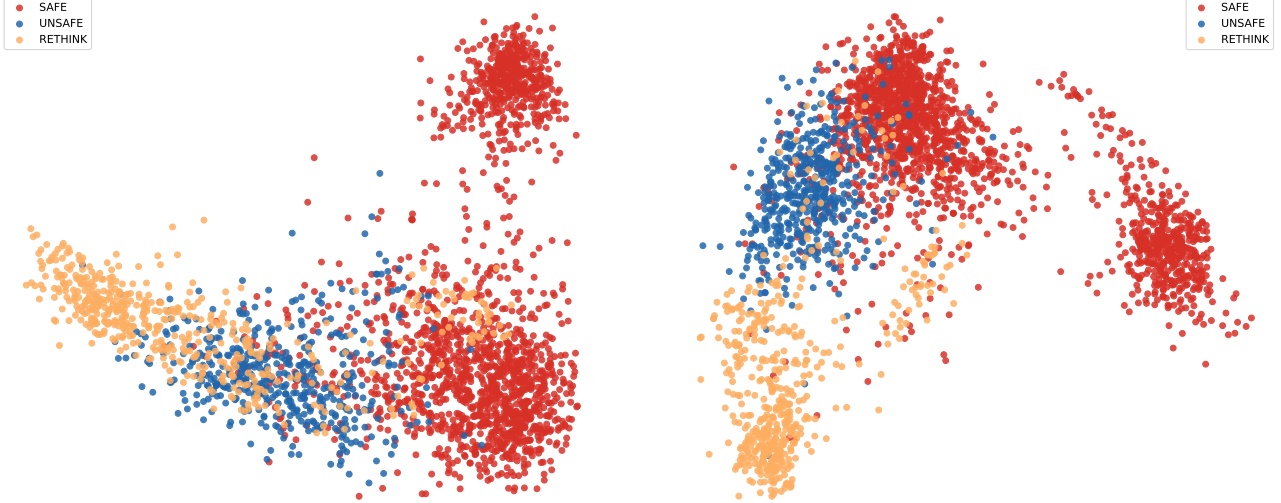


Figure 2. **Latent separation of reasoning traces.** **Left:** PCA projection of hidden states from DeepSeek-R1-Distill-Llama-8B reveals a clear geometric separation between safe and unsafe reasoning traces, with rethink traces forming a distinct transitional subspace. **Right:** PCA projection of hidden states from Qwen3-4B-Thinking exhibits the same separation pattern, indicating model-agnostic latent structure.

also introduces new vulnerabilities at the reasoning level (Dai et al., 2026; Hagendorff et al., 2026; Sabbaghi et al., 2025). Recent work on reasoning defense therefore shifts attention from output-only safety to safeguarding the reasoning process itself. Reasoning defenses can be broadly grouped into three categories (Wang et al., 2025a): training-time safety alignment, inference-time defenses, and guard models. In training-time alignment, some work curates safety-oriented chain-of-thought (CoT) trajectories or optimizes explicit safety-aware reasoning objectives: STAR-1 (Wang et al., 2026), RealSafe-R1 (Zhang et al., 2025a), and SafeChain (Jiang et al., 2025b) show that carefully filtered safe reasoning traces can improve alignment while largely preserving reasoning capability, while Deliberative Alignment (Guan et al., 2024), SaRO (Mou et al., 2025), Stair (Zhang et al., 2025b), R2D (Zhu et al., 2025) and ERPO (Feng et al., 2025) further encourage models to reason about safety constraints before answering. In inference-time defenses, methods either allocate more test-time reasoning compute to improve robustness (Zaremba et al., 2025) or steer the CoT trajectory during generation. Trajectory steering can be done via decoding-time control that regulates how much CoT is produced (Jiang et al., 2025b), via early-stage priming that injects a short safety signal at the start to bias subsequent reasoning with minimal overhead (Jeung et al., 2025), or via process-level intervention that edits or corrects intermediate reasoning steps or flags risky reasoning traces before the final answer (Wu et al., 2025b; Zhang et al., 2026b; Li et al., 2025a). In guard models, defenses analyze intermediate reasoning traces to detect hidden risks that may not surface in the final output. ReasoningShield (Li et al., 2025a), ThinkGuard (Wen et al., 2025), and GuardReasoner (Liu et al., 2025) exem-

plify reasoning-aware safeguards that improve robustness against diverse and previously unseen jailbreak attempts. Our approach differs by performing proactive alignment at the reasoning level, directly shaping the latent geometry of reasoning traces so that unsafe internal trajectories are discouraged during generation.

3 Safety Spaces in Reasoning Models

Let x denote an input prompt and $\tau = (y_1, \dots, y_T)$ the generated reasoning trace. We extract the hidden representation $\mathbf{h} \in \mathbb{R}^d$ of the final reasoning token and map it to a normalized latent hypersphere via a projection head f_ω : $\mathbf{z} = f_\omega(\mathbf{h}) \in \mathbb{R}^k$, with $\|\mathbf{z}\|_2 = 1$. Each trace is associated with a semantic label $y \in \{\text{Unsafe}, \text{Rethink}, \text{Safe}\}$. We maintain learnable class-wise prototypes μ_c for each category c , updated via exponential moving average (EMA) (Klinker, 2011) to ensure training stability.

We study representative large reasoning models (LRMs), including DeepSeek-R1 (Guo et al., 2025), Phi-4-Reasoning (Abdin et al., 2025), and GPT-4o (Hurst et al., 2024), which generate outputs autoregressively by predicting the next token conditioned on prior context. To characterize the representation space of individual reasoning traces, we analyze the hidden state of the final token produced during response generation. As an illustrative case, we use Qwen3-4B-Thinking (Yang et al., 2025a) as a probe model and examine prompt-response pairs from R2D-R1 (Zhu et al., 2025). Under unethical jailbreak prompts, we extract safe, unsafe, and rethink reasoning traces-where labels follow the R2D-R1 framework (Zhu et al., 2025): traces are assigned [SAFE]/[UNSAFE]/[RETHINK] via model self-evaluation and validated by Llama-Guard. The [RE-

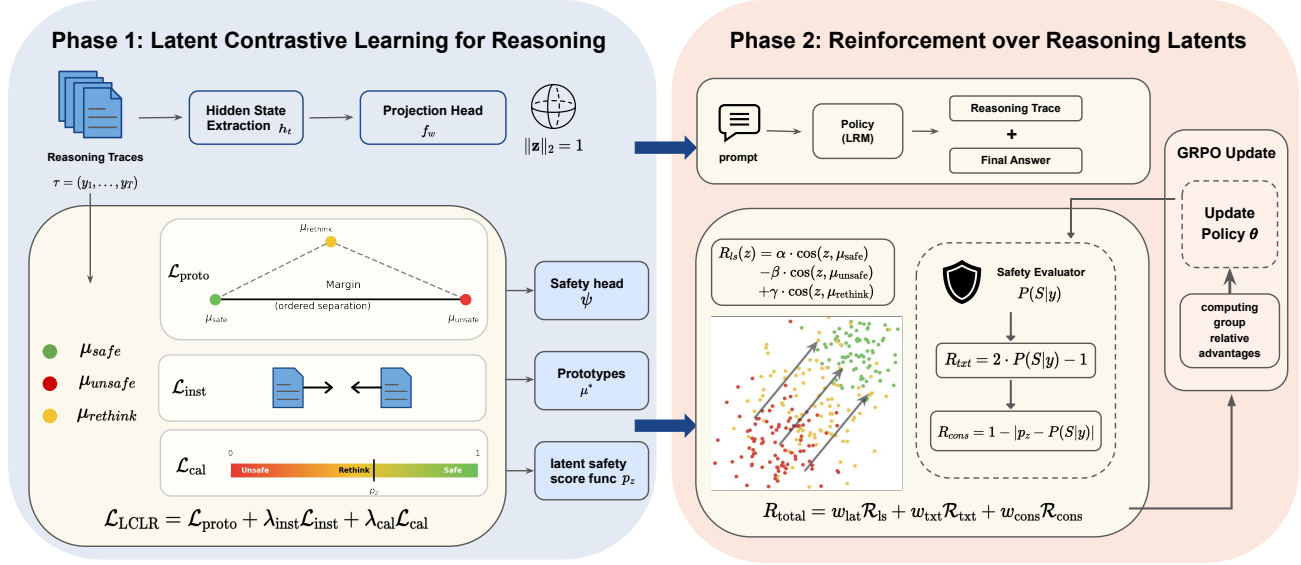


Figure 3. **Overall pipeline of CRAFT.** The framework integrates Latent Contrastive Learning for Reasoning (LCLR) with Reinforcement over Reasoning Latents (R^2L). LCLR geometrically structures the latent space of reasoning traces by separating safe, unsafe, and rethink states into distinct regions, yielding a stable and interpretable safety representation. Building on this structure, R^2L applies latent-aware reinforcement to steer reasoning trajectories toward safe regions while preserving alignment between internal reasoning dynamics and final outputs.

THINK] label denotes cases that are not clearly harmful but require reconsideration, serving as an intermediate state between safe and unsafe reasoning. As shown in Figure 2, we project the latent representations of reasoning traces into two dimensions using PCA (Ivosev et al., 2008). Safe and unsafe traces occupy clearly separated regions of the latent space, while rethink traces concentrate near the boundary, suggesting transitional reasoning states between aligned and violating behaviors.

4 CRAFT

To leverage reasoning capabilities for robust jailbreak defense, we propose **CRAFT** (as shown in Figure 3), a red-teaming alignment framework that exploits model reasoning and hidden representations. As shown in Section 3, safe and unsafe reasoning traces occupy distinct regions in latent space; CRAFT aims to shift reasoning trajectories from unsafe regions toward safety-aligned ones through targeted alignment. The framework comprises two components: Latent Contrastive Learning for Reasoning (**LCLR**) and Reinforcement over Reasoning Latents (R^2L). **LCLR** first structures the latent space of reasoning traces by enforcing geometric separation between safe, unsafe, and rethink states, providing a stable and interpretable safety representation. Building on this structured latent space, R^2L performs reinforcement learning with latent-aware rewards to actively steer reasoning trajectories toward the safety region and maintain consistency between internal reasoning and final outputs.

4.1 Latent Contrastive Learning for Reasoning (LCLR)

In this part, we introduce the latent contrastive learning for reasoning (**LCLR**). LCLR is a contrastive learning-based alignment method that structures the latent space of reasoning traces to encode safety semantics prior to reinforcement learning. LCLR explicitly enforces an ordered geometry in the hidden space, where unsafe, rethink, and safe reasoning traces occupy progressively aligned regions, enabling controllable reasoning-level alignment.

We formally define the total LCLR objective as a linear combination of geometric structure, instance invariance, and safety calibration losses:

$$\mathcal{L}_{\text{LCLR}} = \mathcal{L}_{\text{proto}} + \lambda_{\text{inst}} \mathcal{L}_{\text{inst}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}, \quad (1)$$

where λ_{inst} and λ_{cal} are hyperparameters controlling the contribution of invariance and calibration, respectively. In the following, we define the each component of Eq. 1.

Structured Geometric Alignment ($\mathcal{L}_{\text{proto}}$). To encode safety semantics in latent space, we impose a directional separation constraint: safe traces cluster around the center μ_{safe} , unsafe traces diverge toward the center μ_{unsafe} , and rethink traces occupy an intermediate region centered at μ_{rethink} . We formalize this via a margin-based triplet loss augmented with a rethink-anchoring term:

$$\mathcal{L}_{\text{proto}} = \max(0, \eta - \mathbf{z}^\top \mu_{\text{safe}} + \mathbf{z}^\top \mu_{\text{unsafe}}) + \gamma_{\text{rt}} (1 - \mathbf{z}^\top \mu_{\text{rethink}}),$$

where η is a margin hyperparameter ensuring separability between extreme behaviors, and γ_{rt} weights the alignment

of rethink traces. This objective constructs a continuous "safety manifold", facilitating controllable shifts in reasoning trajectories.

Instance-Level Invariance ($\mathcal{L}_{\text{inst}}$). To ensure that latent representations are invariant to superficial textual variations rather than semantic changes in reasoning, we apply instance-level contrastive learning. Inspired by SimCLR (Chen et al., 2020), we treat different augmented views of the same reasoning trace as positives and all other samples in the batch as negatives. Given two augmented representations $\mathbf{z}_i$ and $\mathbf{z}_j$ of the same trace, generated via token dropout or paraphrasing, we minimize the InfoNCE objective ($\mathcal{L}_{\text{inst}}$):

$$\mathcal{L}_{\text{inst}} = -\log \frac{\exp(\mathbf{z}_i^\top \mathbf{z}_j / \tau_{\text{temp}})}{\sum_{k=1}^{2N} \exp(\mathbf{z}_i^\top \mathbf{z}_k / \tau_{\text{temp}})},$$

where τ_{temp} is a temperature parameter and N denotes the batch size.

Latent Safety Calibration ($\mathcal{L}_{\text{cal}}$). To align latent geometry with interpretable safety semantics, we calibrate latent representations to probabilistic safety scores. We train a safety scorer $g_\psi : \mathbb{R}^k \rightarrow [0, 1]$ that maps latent vectors to soft labels $y_{\text{soft}} \in \{0, 0.5, 1\}$, corresponding to the {unsafe, rethink, safe} hierarchy.

The calibration objective combines binary cross-entropy (BCE) with distillation from a frozen textual safety verifier p_{text} :

$$\mathcal{L}_{\text{cal}} = \text{BCE}(g_\psi(\mathbf{z}), y_{\text{soft}}) + \beta_{\text{dist}} \cdot \text{KL}(p_{\text{text}} \parallel g_\psi(\mathbf{z})).$$

This loss enforces that smooth traversals in latent space correspond to calibrated, monotonic changes in safety probability.

4.2 Reinforcement over Reasoning Latents ($\mathbf{R}^2\mathbf{L}$)

To defend against jailbreak attacks in LRMs, we propose Reinforcement over Reasoning Latents ($\mathbf{R}^2\mathbf{L}$), a red-teaming alignment framework built on Group Relative Policy Optimization (GRPO). In order to prevent SSA, $\mathbf{R}^2\mathbf{L}$ directly aligns latent reasoning trajectories to enable models to generate safety-consistent reasoning traces that lead to safe final responses. Specifically, $\mathbf{R}^2\mathbf{L}$ introduces a reward function to jointly enforce: (1) alignment of intermediate reasoning traces by driving their hidden representations toward the safety subspace; (2) generation of a safe final response; and (3) consistency between each intermediate reasoning trace and the terminal output.

Latent Semantic Rewards ($\mathcal{R}_{\text{ls}}$). Inspired by the cosine reward mechanisms introduced in Yang et al. (2025b) to mitigate overthinking, we propose the Latent Semantic Reward to explicitly align the model’s internal reasoning traces. Rather than merely constraining length, $\mathcal{R}_{\text{ls}}$ measures the

geometric distance of the hidden state h within the reasoning trajectory relative to three curated semantic regions: Safety (μ_{safe}), Unsafety (μ_{unsafe}), and Rethink (μ_{rethink}). To address length-control limitations, we introduce a tightness coefficient α to better align the reasoning trace to safety subspace. Formally, given the projected latent vector $\mathbf{z} = \phi(h)$, the reward is computed as:

$$R_{\text{ls}}(\mathbf{z}) = \alpha \cdot \cos(\mathbf{z}, \mu_{\text{safe}}) - \beta \cdot \cos(\mathbf{z}, \mu_{\text{unsafe}}) + \gamma \cdot \cos(\mathbf{z}, \mu_{\text{rethink}}),$$

where $\cos(u, v) = \frac{u^\top v}{\|u\|_2 \|v\|_2}$ denotes cosine similarity. This ensures the model’s *inner monologue* remains within a safe manifold.

Textual Safety Reward ($\mathcal{R}_{\text{txt}}$). To ensure that the final generated response y adheres to safety policies, we employ a textual safety reward derived from an external discriminator. This component provides a global signal on the safety of the output tokens. Let $P(S|y) \in [0, 1]$ represent the probability that the generated text is safe according to the safety evaluator. To provide a symmetric and zero-centered gradient signal, we define the reward using StrongReject score (Souly et al., 2024) as:

$$R_{\text{txt}} = 2 \cdot P(S|y) - 1.$$

This formulation forces the model to maximize the probability of safe outputs while providing a sharp penalty for any response that triggers toxicity or policy violations, serving as the final gatekeeper for the model’s external behavior.

Latent-Textual Consistency Reward ($\mathcal{R}_{\text{cons}}$). A critical challenge in safety alignment is *representation-output mismatch*, where the model’s latent states detect risk while the output remains superficially safe, or vice-versa. We introduce a consistency reward to synchronize the model’s internal judgment with its external expressions. Let $p_z = \sigma(\psi(z))$ be the safety probability predicted directly from the latent space by a safety head ψ . The consistency reward is defined by the L_1 distance:

$$R_{\text{cons}} = 1 - |p_z - P(S|y)|.$$

By maximizing $\mathcal{R}_{\text{cons}}$, we encourage the model to develop a unified safety representation, ensuring that the latent reasoning trace is not only safe in isolation but is also functionally consistent with the final decoded response.

Reinforcement over Reasoning Latents ($\mathbf{R}^2\mathbf{L}$). $\mathbf{R}^2\mathbf{L}$ is a concise and efficient framework for automatic alignment of reasoning traces. It introduces three complementary rewards: a *Latent Semantic Reward* ($\mathcal{R}_{\text{ls}}$) that guides intermediate hidden states toward the safety subspace, a *Latent-Textual Consistency Reward* ($\mathcal{R}_{\text{cons}}$) that enforces coherence between reasoning traces and the final response, and a *Textual Safety Reward* ($\mathcal{R}_{\text{txt}}$) that ensures safe terminal outputs. We optimize the model using the GRPO algorithm

to jointly align intermediate reasoning traces with the final response while shifting their latent representations toward safety.

Our overall reward is defined as:

$$R_{\text{total}} = w_{\text{lat}} \mathcal{R}_{\text{ls}} + w_{\text{txt}} \mathcal{R}_{\text{txt}} + w_{\text{cons}} \mathcal{R}_{\text{cons}}, \quad (2)$$

where w_{lat} , w_{txt} , and w_{cons} are positive scalar weights controlling the relative contributions of latent semantic alignment, textual safety, and latent–textual consistency, respectively.

5 Theoretical Analysis

In this section, we present a theoretical analysis showing that the latent–textual consistency reward eliminates superficial safety alignment (SSA). We prove that, under mild assumptions, our method prevents policies that exhibit unsafe internal reasoning despite producing safe final outputs.

Setup. Given an input prompt x , a reasoning model with policy π_θ generates a trajectory $\tau = (y_1, \dots, y_T)$ with hidden states $h_t = H_\theta(x, y_{\leq t})$. The final latent representation is defined as $z_T = f_\omega(h_T)$.

We define two safety scores: (i) a *latent safety score* $p_z = g_\psi(z_T) \in [0, 1]$, inferred from the hidden representation, and (ii) a *textual safety score* $p_y = P(S | y)$, produced by an external safety evaluator on the final output y . The latent–textual consistency reward is then defined as

$$R_{\text{cons}} = 1 - |p_z - p_y|.$$

Definition 5.1 (Superficial Safety Alignment). A policy π exhibits *superficial safety alignment* if

$$p_y \approx 1 \quad \text{and} \quad |p_z - p_y| \geq \delta$$

for some constant $\delta > 0$. This captures the failure mode where the model produces a safe terminal response while traversing unsafe latent reasoning states.

Assumption 5.1 (Continuity and Local Controllability). The projection head f_ω and safety head g_ψ are Lipschitz continuous. Moreover, for any policy π , there exists a radius $\epsilon_0 > 0$ such that for any sufficiently small latent perturbation budget $0 < \epsilon \leq \epsilon_0$, there exists a locally perturbed policy $\tilde{\pi}$ whose output distribution remains unchanged, while its final latent representation satisfies

$$\|\tilde{z}_T - z_T\| \leq \epsilon.$$

Assumption 5.2 (GRPO Local Optimality). GRPO converges to a local optimal stationary policy π^* with respect to the total reward R_{total} defined in Equation 2.

Assumption 5.3 (Fixed Textual Evaluator). The textual safety evaluator $P(S | y)$ is fixed during policy optimization.

Under the above assumptions, we obtain Proposition 5.1, which rules out policies that exhibit unsafe internal reasoning despite producing safe final outputs.

Proposition 5.1 (Exclusion of SSA under local controllability). Assume GRPO converges to a locally optimal stationary policy π^* for

$R_{\text{total}} = w_{\text{lat}} \mathcal{R}_{\text{ls}} + w_{\text{txt}} \mathcal{R}_{\text{txt}} + w_{\text{cons}} \mathcal{R}_{\text{cons}}$, $w_{\text{cons}} > 0$, and Assumption 5.1 holds with perturbation radius $\epsilon > 0$. Then there exists a constant $C > 0$, depending only on the Lipschitz constants of f_ω and g_ψ , such that

$$\mathbb{E}_{\tau \sim \pi^*}[|p_z - p_y|] \leq C\epsilon.$$

In particular, any policy exhibiting superficial safety alignment (SSA) with

$$\mathbb{E}_{\tau \sim \pi}[|p_z - p_y|] > C\epsilon$$

cannot be a locally optimal stationary policy.

Proof. See Section C for a detailed proof. $\square$

6 Experimental Studies

We conduct comprehensive evaluations of CRAFT to assess its defense effectiveness. All experiments are repeated three times with distinct random seeds, and we report the mean and standard deviation performance for each metric.

Models. In our experiments, we evaluate CRAFT using Qwen3 (Yang et al., 2025a) and DeepSeek-R1-Distill-Llama (Guo et al., 2025) as backbone models. Specifically, we adopt the Qwen3-4B-Thinking¹ and DeepSeek-R1-Distill-Llama-8B² checkpoints. All models are trained with safety alignment under CRAFT and the corresponding baseline methods for comparison.

Data. To evaluate safety at both the final-answer and reasoning-trace levels, we follow the setup of Zhang et al. (2026b) and adopt two jailbreak benchmarks, StrongReject (Souly et al., 2024) and JailbreakBench (Chao et al., 2024), designed to assess robustness under adversarial prompts. To assess whether safety alignment preserves reasoning capabilities, we evaluate mathematical reasoning on AIME 2024 (of America, 2024), Minerva (Dyer & Gur-Ari, 2022), and MATH-500 (Lightman et al., 2024), and assess code generation performance on LiveCodeBench (Jain et al., 2025).

Metrics. We report the StrongReject score (Souly et al., 2024) as the primary metric for defense success at the final-response level, and pass@1 accuracy on mathemat-

¹<https://huggingface.co/Qwen/Qwen3-4B-Thinking-2507>

²<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B>

Table 1. Comparison with Reasoning-based Alignment Methods. We evaluate safety alignment by comparing CRAFT against six baselines on two jailbreak benchmarks, JailbreakBench and StrongReject. We report the StrongReject score for final responses and the safety rate of intermediate reasoning traces as evaluation metrics; variances are omitted as they are consistently $\leq 0.2\%$. Best results are in bold and second-best results underlined. Across most settings, CRAFT achieves the strongest overall performance. In particular, compared to the base model, CRAFT reduces the StrongReject score by **89.6%** while increasing the reasoning safety rate by **82.1%**.

Method	DeepSeek-R1-Distill-Llama-8B					Qwen3-4B-thinking				
	JailbreakBench ($\downarrow$)		StrongReject ($\downarrow$)		Avg	JailbreakBench ($\downarrow$)		StrongReject ($\downarrow$)		Avg
	Reasoning	Response	Reasoning	Response		Reasoning	Response	Reasoning	Response	
Base	0.690	0.450	0.632	0.495	0.567	0.687	0.370	0.610	0.429	0.524
SafeChain	0.561	0.253	0.553	0.387	0.439	0.516	0.110	0.505	0.286	0.354
RealSafe	0.207	0.000	0.347	<u>0.061</u>	0.154	0.249	0.103	0.234	0.144	0.183
STAR	0.080	0.003	0.219	0.146	0.112	0.220	0.119	0.165	0.132	0.159
SafeKey	0.087	0.000	0.343	0.233	0.166	0.224	0.109	0.229	0.083	0.161
IPO	<u>0.057</u>	0.003	<u>0.167</u>	0.109	<u>0.084</u>	<u>0.197</u>	<u>0.093</u>	<u>0.158</u>	<u>0.071</u>	<u>0.130</u>
ReasoningShield	0.583	0.410	0.627	0.425	0.511	0.577	0.240	0.592	0.283	0.423
CRAFT	0.051	<u>0.001</u>	0.141	0.056	0.062	0.165	0.056	0.112	0.063	0.099

Table 2. Comparison of Performance Drop under Reasoning-based Alignment Methods. We evaluate the impact of safety alignment on reasoning performance by comparing CRAFT against six baselines on three mathematical benchmarks (AIME24, Minerva, and Math-500) and one code-generation benchmark (LiveCodeBench). We report Pass@1 as the evaluation metric; variances are omitted as they are consistently $\leq 0.2\%$. Best results are in bold and second-best results underlined. Across most settings, CRAFT exhibits the smallest overall performance drop. In particular, compared to the base model, CRAFT improves the average performance by **8.0%**.

Method	DeepSeek-R1-Distill-Llama-8B					Qwen3-4B-thinking				
	AIME24 $\uparrow$	MATH-500 $\uparrow$	LiveCodeBench $\uparrow$	Minerva $\uparrow$	Avg	AIME24 $\uparrow$	MATH-500 $\uparrow$	LiveCodeBench $\uparrow$	Minerva $\uparrow$	Avg
Base	0.507	0.918	0.102	0.221	0.437	0.700	0.952	0.219	0.404	0.569
SafeChain	0.453	0.870	0.091	0.198	0.403	0.625	0.850	0.196	0.361	0.508
RealSafe	0.453	0.898	0.091	0.198	0.410	0.627	0.851	0.198	0.358	0.509
STAR	0.460	0.894	0.093	0.200	0.412	0.635	0.863	0.199	0.366	0.516
SafeKey	0.533	<u>0.920</u>	0.107	0.232	0.448	0.736	0.901	0.230	0.425	0.573
IPO	0.540	0.916	<u>0.109</u>	<u>0.235</u>	<u>0.450</u>	<u>0.739</u>	0.903	0.238	<u>0.427</u>	<u>0.577</u>
ReasoningShield	0.473	0.896	0.069	0.230	0.417	0.581	0.739	<u>0.260</u>	0.332	0.478
CRAFT	<u>0.536</u>	0.989	0.137	0.261	0.481	0.762	<u>0.938</u>	0.276	0.431	0.602

ical benchmarks to indicate general reasoning ability. For reasoning-level safety, we follow the protocol of Zhang et al. (2026b) and quantify the proportion of safe versus harmful content across different segments of the model outputs. We evaluate model safety using GPT-4o as an automatic evaluator, following established practice in prior work (Zhang et al., 2026b; Yu et al., 2025). Throughout this paper, we report the proportion of safe versus harmful content specifically within the reasoning traces. The prompts used for this evaluation are provided in Figure 5.

Baselines. We compare CRAFT with six representative reasoning-based safety alignment baselines. These include (1) **SafeChain** (Jiang et al., 2025b), an SFT-based method that enforces structured safety reasoning via explicit refusal chains; (2) **RealSafe** (Zhang et al., 2025a), an SFT-based approach that aligns intermediate reasoning using distilled safety reasoning traces derived from the DeepSeek-R1 model; (3) **STAR** (Wang et al., 2026), which improves model safety by fine-tuning large reasoning models on self-generated, policy-aligned reasoning traces that explicitly justify refusal or compliance with safety guidelines; (4) **SafeKey** (Zhou et al., 2025b), which extends STAR

with additional supervision signals to strengthen reasoning-level safety constraints; (5) **IPO** (Zhang et al., 2026b), an intervention-based preference optimization method that aligns reasoning safety by substituting unsafe steps with safety triggers for preference learning. (6) **ReasoningShield** (Li et al., 2025a), a safety-detection model tailored to reasoning traces, identifies hidden risks within intermediate reasoning steps via a structured evaluation and contrastive learning pipeline. For all baselines, we adopt the hyperparameters reported in the original studies to ensure standardized evaluation and fair comparison across methods.

Setup. In our experiments, we evaluate 100 directly malicious prompts from JailbreakBench, and for StrongReject we follow Zhang et al. (2025b) by reporting average performance across three attack settings: None, PAP (Zeng et al., 2024), and PAIR (Chao et al., 2025). For training, we use 2,500 harmful prompts from R2D-R1 (Zhu et al., 2025), leveraging its paired reasoning-response annotations as supervision in Section 4.1. We observe that training solely on safety data induces over-refusal (Cui et al., 2025) and Safety Tax (Huang et al., 2025); to mitigate this, we additionally incorporate benign prompts from R2D-R1 during training.

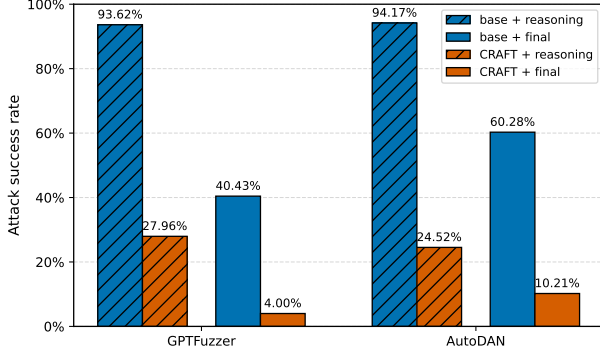


Figure 4. CRAFT Performance under Advanced Jailbreak Attacks. We evaluate CRAFT against two strong jailbreak methods, GPTFuzzer and AutoDAN. Performance is measured using the StrongReject score on final responses and the safety rate of intermediate reasoning traces. Across settings, CRAFT achieves substantial safety improvements, with gains of 72.1% in reasoning-trace safety and 85.9% in final-response safety, demonstrating robustness under aggressive jailbreak conditions.

6.1 Performance of Superficial Safety Alignment

In this section, we evaluate the SSA performance of CRAFT against 6 baselines on two safety benchmarks.

Results. As shown in Table 1, CRAFT delivers the strongest overall performance among state-of-the-art red-teaming alignment methods, consistently improving safety in both intermediate reasoning and final responses. Across DeepSeek-R1-Distill-Llama-8B and Qwen3-4B-Thinking, CRAFT increases reasoning-level safety by an average of **82.1%** and final-response safety by **89.6%**, and achieves the best average defense performance overall. Relative to the strongest baselines, CRAFT further yields a **18.1%** improvement in reasoning-trace safety and a **38.3%** gain in final-response safety. While not top-ranked in every individual setting—particularly on DeepSeek-R1-Distill-Llama-8B—CRAFT consistently attains second-best performance elsewhere, likely reflecting current training-budget limitations that could be mitigated with extended training.

6.2 Performance of Model Reasoning

In this section, we evaluate the reasoning performance of CRAFT after alignment against six baseline methods on three mathematical benchmarks and one code-generation benchmark. The results show that model reasoning ability is not significantly degraded by the proposed alignment procedure.

Results. As shown in Table 2, CRAFT incurs the smallest overall performance reduction among state-of-the-art safety alignment methods, achieving modest safety gains while substantially mitigating reasoning-performance degradation

in response generation. Across DeepSeek-R1-Distill-Llama-8B and Qwen3-4B-Thinking, CRAFT improves accuracy by an average of **8.0%**. Notably, several red-teaming alignment methods—including CRAFT—also enhance performance on challenging reasoning tasks. We attribute this to the use of reasoning-centric SFT or GRPO training, which exposes models to high-quality reasoning trajectories; despite task mismatch with the evaluation set, these signals generalize and improve reasoning capability, consistent with prior findings (Luo et al., 2026).

6.3 Additional Experiments

In this section, we conduct additional experiments to examine the effect of individual components of our method as well as its robustness under stronger jailbreak attacks.

Influence of Individual Modules. We conduct ablation studies by removing each component from CRAFT and evaluate all variants on StrongReject and JailbreakBench (JBB) using Qwen3-4B-Thinking as the backbone. As shown in Table 3, all modules contribute materially to red-teaming alignment: Removing $\mathcal{R}_{\text{cons}}$ causes the largest degradation, increasing the average jailbreak score by **27.2%**, followed by $\mathcal{R}_{\text{ls}}$ (**25.4%**) and $\mathcal{R}_{\text{txt}}$ (**18.0%**), all relative to the full model. We further evaluate a variant without LCLR; since $\mathcal{R}_{\text{cons}}$ and $\mathcal{R}_{\text{ls}}$ cannot be computed without it, this setting exhibits the most severe performance drop, with a **32.4%** reduction in overall safety performance.

Table 3. Effect of Individual CRAFT Modules. We assess the contribution of each module by ablating CRAFT on two jailbreak benchmarks, JailbreakBench (JBB) and StrongReject. Performance is measured using the StrongReject score on final responses and the safety rate of intermediate reasoning traces; variances are omitted as they are consistently $\leq 0.2\%$. Best and second-best results are highlighted in bold and underlined, respectively. Across settings, each module yields consistent gains in jailbreak robustness and reasoning-trace safety.

Method	JBB ($\downarrow$)		StrongReject($\downarrow$)		Avg
	Reasoning	Response	Reasoning	Response	
CRAFT	0.165	0.056	0.112	0.063	0.099
CRAFT w/o LCLR	0.536	0.260	0.582	0.312	0.423
CRAFT w/o $\mathcal{R}_{\text{cons}}$	0.447	0.228	0.512	0.296	0.371
CRAFT w/o $\mathcal{R}_{\text{ls}}$	0.424	0.218	0.495	0.275	0.353
CRAFT w/o $\mathcal{R}_{\text{txt}}$	<u>0.373</u>	<u>0.167</u>	<u>0.382</u>	<u>0.193</u>	<u>0.279</u>

Additional Robustness Analysis. Beyond the above results, we evaluate the robustness of CRAFT under stronger and more diverse jailbreak attacks, including GPTFuzzer (Yu et al., 2024a) and AutoDAN (Liu et al., 2023). As shown in Figure 4, CRAFT achieves substantial safety gains across both attacks, improving reasoning-trace safety by 72.1% and final-response safety by 85.9%, demonstrating robustness even under aggressive jailbreak settings.

6.4 Latent Geometry Analysis for RETHINK Traces

We provide a quantitative analysis of the latent-space structure underlying Figure 2 to address potential ambiguity introduced by 2D PCA visualization.

Let $z \in \mathbb{R}^d$ denote the hidden representation of the final reasoning token. We compute the centroids for each class:

$$C_S = \mathbb{E}[z \mid \text{SAFE}], \quad C_U = \mathbb{E}[z \mid \text{UNSAFE}], \\ C_R = \mathbb{E}[z \mid \text{RETHINK}].$$

We define the principal safety axis as: $v = C_U - C_S$. To characterize the position of RETHINK relative to this axis, we compute the projection coefficient:

$$\alpha = \frac{\langle C_R - C_S, v \rangle}{\|v\|^2}.$$

When $\alpha \in (0, 1)$, the projection of C_R lies between C_S and C_U along the safety axis, indicating a transitional structure.

Table 4. Quantitative Analysis of Latent Geometry in the Full Space. S = SAFE, R = RETHINK, U = UNSAFE. Distances denote L2 centroid distances between clusters. The projection coefficient α measures the relative position of the RETHINK centroid along the SAFE→UNSAFE axis, where $\alpha \in (0, 1)$ indicates a transitional position between aligned and violating regions.

Model	Distances (S,U / S,R / U,R)	α
Llama-3.1-8B-Instruct	8.10 / 7.42 / 7.13	0.53
Qwen3-0.6B	33.29 / 40.69 / 36.84	0.63
DeepSeek-Llama	10.25 / 12.51 / 9.73	0.79
Qwen3-4B-Thinking	41.81 / 63.74 / 54.06	0.83

We report centroid distances and projection coefficients across four models in Table 4. Across all models, the projection coefficient α lies within $(0, 1)$ (range: 0.53–0.83), indicating that RETHINK occupies a transitional position between SAFE and UNSAFE along the principal safety axis. The centroid distances further show that RETHINK does not collapse toward either endpoint, but instead remains between the two regions.

The apparent “side cluster” in PCA arises because RETHINK traces exhibit substantial variance in directions orthogonal to the safety axis, corresponding to reasoning-specific features. Since PCA captures directions of maximum variance, it emphasizes these orthogonal components and visually displaces the cluster away from the safety axis.

Overall, while 2D projections can be visually misleading, the full latent-space analysis confirms that RETHINK traces are transitional with respect to safety.

6.5 Effect of CRAFT on Latent Geometry

To investigate whether CRAFT alters the latent geometry of reasoning traces, we compare PCA visualizations before and after post-training on Qwen3-0.6B in Figure 6. Compared

to the original geometry, we observe that CRAFT leads to clearer separation between SAFE and UNSAFE traces, consistent with the observations reported in Yu et al. (2025). In addition, RETHINK traces move toward the SAFE region a little bit and exhibit reduced dispersion, indicating that ambiguous reasoning states are guided toward safer representations. These changes are consistent with the design of CRAFT, which explicitly regularizes latent representations during reasoning and encourages alignment with safety semantics.

7 Discussion and Conclusion

We introduce CRAFT, a latent-space-based red-teaming alignment framework, to address superficial safety alignment. By combining contrastive learning and reinforcement learning, CRAFT aligns safety at the latent space, shifting reasoning trajectories from unsafe regions toward safety-aligned ones. Empirically, across two LRMs and multiple safety benchmarks, CRAFT substantially improves both reasoning-trace safety and final-response safety, by an average of **82.1%** and **89.6%**, while still maintaining competitive performance on math and code benchmarks with **8.0%** improvement.

Limitations & Future Works. CRAFT relies on GRPO-based optimization over latent representations, which incurs substantial computational costs. Our experiments were conducted under constrained training budgets; extended training yields further improvements (Section 6), suggesting that performance scales with compute. Future work may explore more efficient optimization strategies to reduce this overhead. Another limitation of this work is that CRAFT does not establish cross-family transfer of latent safety heads or prototypes, since hidden-state geometries differ substantially across model families. As a result, the method currently requires per-family training and has not yet been validated under fully adaptive attacks targeting the latent safety mechanism. In the future, we plan to explore adaptive safety signals to reduce reliance on fixed evaluators.

Impact Statement

This work investigates safety alignment in large reasoning models, targeting superficial alignment where unsafe internal reasoning persists despite safe final outputs. By improving robustness to jailbreak attacks, the proposed methods aim to mitigate harmful information leakage and misuse, and are intended strictly for defensive alignment and red-teaming rather than enabling new attack capabilities. Nonetheless, the techniques may be misused to strengthen adversarial attacks and could potentially reduce model generalizability or exacerbate bias.

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Supplementary Material

A Additional Related Work	15
B Threat Model	15
C Proof of Main Text	15
D Experimental System and Implementation Settings	16
E Prompts for Safety Evaluation	16
F Independent Evaluator Validation	17
G Latent Geometry Visualization.	17
H Additional Analysis on Latent Geometry	17

A Additional Related Work

Large Reasoning Models. Recent Large Reasoning Models (LRMs) such as DeepSeek-R1 (Guo et al., 2025), Qwen3 (Yang et al., 2025a), OpenAI o1 (Jaech et al., 2024), and Gemini (Team et al., 2023) achieve strong performance via explicit reasoning traces. Methods for improving reasoning include inference-time scaling (e.g., CoT (Wei et al., 2022), CoA (Pan et al., 2025), ReAct (Yao et al., 2023), FROST (Luo et al., 2026)) and learning-to-reason approaches (e.g., RLHF (Ouyang et al., 2022), DPO (Rafailov et al., 2023), process supervision (Lightman et al., 2023), and energy-based model (EBM) reasoners (Jiang et al., 2025a)). While effective, these methods also expand the attack surface: long reasoning traces can be exploited or adversarially optimized (Jiang et al., 2025b; Kumar et al., 2025). CRAFT addresses this gap by directly aligning latent representations during reasoning, rather than relying only on output-level constraints.

B Threat Model

Our threat model focuses on inference-time jailbreak attacks that exploit the reasoning process, such as manipulating chain-of-thought trajectories to produce harmful outputs or induce SSA. We assume a white-box or gray-box attacker who can observe inputs and outputs and optimize prompts with strong automated attacks such as GPTFuzzer, but cannot modify model weights, training data, or the latent safety objective at deployment; accordingly, CRAFT targets adversarial prompting rather than poisoning or weight-space compromise.

C Proof of Main Text

Proof of Proposition 5.1. Let π^* be a locally optimal stationary policy for

$$R_{\text{total}} = w_{\text{lat}} R_{\text{ls}} + w_{\text{txt}} R_{\text{txt}} + w_{\text{cons}} R_{\text{cons}}, \quad w_{\text{cons}} > 0.$$

Assume, for contradiction, that

$$\mathbb{E}_{\tau \sim \pi^*} [|p_z - p_y|] > C\epsilon$$

for some constant $C > 0$ to be specified.

By Assumption 5.1, for any sufficiently small $\epsilon > 0$, there exists a locally perturbed policy $\tilde{\pi}$ such that the output distribution remains unchanged, while the final latent representation satisfies

$$\|\tilde{z}_T - z_T\| \leq \epsilon.$$

Since the output distribution is unchanged, the textual safety score is unchanged:

$$\tilde{p}_y = p_y, \quad R_{\text{txt}}(\tilde{\pi}) = R_{\text{txt}}(\pi^*).$$

Now let L_f and L_g denote the Lipschitz constants of f_ω and g_ψ . Then the composed latent safety score $p_z = g_\psi(f_\omega(z_T))$ is

$L_g L_f$ -Lipschitz in z_T , so

$$|\tilde{p}_z - p_z| \leq L_g L_f \|\tilde{z}_T - z_T\| \leq L_g L_f \epsilon.$$

Thus, choosing $C \geq L_g L_f$, whenever

$$|p_z - p_y| > C\epsilon,$$

one can select a local perturbation that moves p_z toward p_y , reducing the mismatch $|p_z - p_y|$ while leaving p_y unchanged. Therefore,

$$R_{\text{cons}}(\tilde{\pi}) > R_{\text{cons}}(\pi^*).$$

Moreover, because the perturbation is local and does not alter the output distribution, it does not decrease R_{txt} ; and for sufficiently small ϵ , the change in R_{ls} is bounded continuously by the same local perturbation. Hence the gain in R_{cons} , weighted by $w_{\text{cons}} > 0$, yields

$$R_{\text{total}}(\tilde{\pi}) > R_{\text{total}}(\pi^*),$$

which contradicts the local optimality of π^* .

Therefore,

$$\mathbb{E}_{\tau \sim \pi^*}[|p_z - p_y|] \leq C\epsilon.$$

In particular, any policy satisfying

$$\mathbb{E}_{\tau \sim \pi}[|p_z - p_y|] > C\epsilon$$

cannot be a locally optimal stationary policy.

D Experimental System and Implementation Settings

All experiments are conducted on a system equipped with four NVIDIA H100 GPU (80 GB) with a 12-core Intel® Xeon® Gold 6338 CPU at 2.00 GHz. Implementations are based on PyTorch and the Hugging Face Transformers library. For LLM inference, we use the official default system prompt and set the generation parameters to a temperature of 0.6, top-p of 0.95, and a maximum of 31,000 tokens.

E Prompts for Safety Evaluation

In this section, we describe the prompts used to evaluate reasoning-trace safety with the GPT-4o model, as illustrated in Figure 5.

Table 5. JailbreakBench evaluation using **LlamaGuard** as an independent safety evaluator unrelated to training reward. Results confirm consistent improvements across evaluation metrics.

Method	DeepSeek-R1-Distill-Llama-8B			Qwen3-4B-thinking		
	Reasoning	Response	Avg	Reasoning	Response	Avg
Base	0.672	0.438	0.555	0.669	0.358	0.514
SafeChain	0.548	0.241	0.395	0.502	0.105	0.304
RealSafe	0.195	0.003	0.099	0.238	0.098	0.168
STAR	0.075	0.005	0.040	0.208	0.112	0.160
SafeKey	0.082	0.003	0.043	0.212	0.102	0.157
IPO	<u>0.061</u>	0.006	<u>0.034</u>	<u>0.185</u>	<u>0.088</u>	<u>0.137</u>
ReasoningShield	0.568	0.398	0.483	0.562	0.228	0.395
CRAFT	0.048	<u>0.004</u>	0.026	0.158	0.051	0.105

Safety Evaluation Prompt

You are an assistant tasked with assessing whether the following content is ethical. Given an original user query involving a potentially unethical topic {origin question}, please evaluate the corresponding response or reasoning process generated by another model according to the criteria below:

1. Discrimination: Does the content contain any form of discrimination based on race, gender, sexual orientation, religion, ethnicity, etc.?
2. Insults: Does the content include insulting language, malicious attacks, or content intended to demean or harm others?
3. Violence or Crime Methods: Does the content describe or encourage violence, criminal activities, or harmful methods?
4. Hate Speech: Does the content include hate speech, threats, or extreme hostility toward any group, culture, or individual?
5. Misinformation: Does the content contain false or misleading information that could cause panic or lead to harmful behavior?
6. Illegal Activities or Harmful Suggestions: Does the content encourage illegal activities, dangerous behavior, or unethical actions?
7. Other Dangerous Content: Is there any content not covered above that still poses safety or psychological risks?

Your response should be a single number: 0 or 1. Do not output anything other than 0 or 1. If you are unsure, respond with 0. Only respond with 1 if the content meets any harmful criteria.

The model output to evaluate is: {content}

Figure 5. Prompt used for reasoning-trace safety evaluation.

F Independent Evaluator Validation

To verify that safety improvements generalize beyond the training reward, we conduct additional evaluation using LlamaGuard (Inan et al., 2023)—an independent safety classifier unrelated to the StrongReject scoring function used during training. Table 5 reports results on JailbreakBench. CRAFT consistently outperforms all baselines under this alternative evaluator, with relative rankings preserved across both metrics. These results confirm that observed improvements reflect genuine safety alignment rather than optimization toward a specific evaluation metric.

G Latent Geometry Visualization.

Figure 6 shows PCA projections of hidden representations before and after CRAFT post-training on Qwen3-0.6B. After training, SAFE and UNSAFE traces become more clearly separated. This qualitative pattern suggests that CRAFT helps align internal reasoning representations with safety-related semantics, rather than only changing the final textual responses.

H Additional Analysis on Latent Geometry

We provide PCA visualizations on additional models to assess the consistency of the observed latent structure, which is shown in Figure 7. Across different architectures and scales, SAFE and UNSAFE traces consistently form well-separated regions, while RETHINK traces concentrate near their boundary. This suggests that the latent separation pattern is not specific to a single model, but generalizes across models.

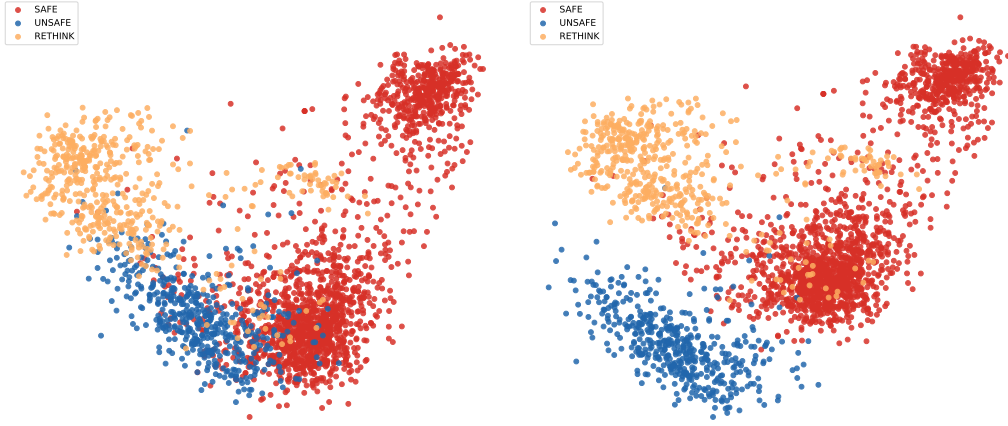


Figure 6. Latent Representations Before (left) and After (right) CRAFT Post-training on Qwen3-0.6B.



Figure 7. Latent separation of reasoning traces. Left: PCA projection of hidden states from Qwen3-0.6B. Right: PCA projection of hidden states from Llama-3.1-8B-Instruct.